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Investment funds management strategy based on polynomial regression in machine learning

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Abstract

This paper presents the results of an investment strategy simulation. The strategy is based on common regression models in a time series, which yields the decision. A simple polynomial regression was the basic method used to achieve short-term value forecasts in the time series. Base params (number of steps in the past and a degree of a polynomial) were set based on a machine learning algorithm. The strategy is improved with some additional original (constitutes by authors) parameters because using only the regression proved to be completely ineffective. Financial markets with bidirectional transactions (long and short transactions), as well as only long transaction markets, were both taken under research.

Keywords: time series, prediction, finance markets, simulation, algo trading, machine learning

1 Introduction.

To document the effectiveness of the strategy, the authors carried out some simulations on different financial markets; often on markets with entirely different characteristics and variability over time. Forex (Foreign Exchange) is one of today's most popular remote access markets to invest on. It's enabling quick and high returns on invested funds. Unfortunately it is also a high risk market, where players can lose their assets very quickly. With Forex it is very difficult to provide an accurate prediction on currency pairs, this makes it difficult to trade safely. Prediction (prediction estimation before validation) should be seen as a method to react to changes in the time series. First attempts to simulate, as covered in this article, were taken on a EUR/USD currency pair as the test data to represent the financial market. This currency pair currently has the largest volume and is the most chosen financial instrument. Tests were taken on hourly OHLC bars for the period between 2200 on 17th January 2012 and 1600 on 26th June 2013, and the source of a data was bossafx.pl trading service. The OHLC chart is a representation of the financial market data as a time series where each time period is described by four prices: O - open price of a period, H - highest

price in a period, L - lowest price in a period, C - close price in a period, and trading volume.

In this paper, the efficacy of the strategy for a completely different financial market such as Investment Funds was also studied. The time series for investment funds has a completely different characteristic; they are focused on long-term investment (months or even years). Authors present a completely different approach based on the same concept of polynomial regression. Data was taken from bossafund.pl online brokerage house. The data includes values for individual funds which depend on the funds from the previous day, or a few days earlier. These values (for most of funds) are updated on the portal every day from 9am local time. Because the funds are listed on various stock exchanges, the data will not be covered continuously on all of the markets. To keep continuity of the data and avoid any issues triggered by gaps in the time series, missing data was populated using the last given value of asset.

Strategies based on the linear or the higher degree polynomial regression, as well as a multiregression, are well known methods and have a long history. Other authors, like Draper, Fujimoto, Elder and Breiman [5, 6, 4, 7], are directly or indirectly using those methods in their research. The first author of this paper has previously published papers [8, 9] in this area. These methods used in their original form are not often effective, sometimes even completely useless. Almost always they can give good results in a short, closed time period. It's a result of a natural value found by machine learning (local optimization). Of course it could be different in a different time window of the time series. Fusion with the other prediction methods can significantly improve efficacy; In particular, good results could be achieved from a fusion of regression methods with various methods based on simple investment rules, like pattern recognition, moving average, pivot points, stop loss and take profit thresholds, and patterns from the traditional technical analysis. There is a wide range of literature concerned with simple rules in prediction on financial markets. First attempts to use these methods were taken in the 1980's, and continues to current day [10, 11, 12, 13, 14]. The special interest we can see in comparing risks Forex approach [1, 2] and more enhanced methods [3, 4] with slowly progress.

The authors aim is to prove that the combination of the two approaches allows us to achieve a better strategy and to find authors original solution with good results of simulation.

2 Characteristics of Investment Strategy

The first of the strategies for consideration (called reg1t), is based on a simple linear regression, attempting to predict time series value change in next step ($i+1$). The example results of this strategy are represented by figure 1.

Figure 1 shows actual time series values and some linear model built on $k=11$ steps back. Approximation with the linear model was taken on a segment of a time series between $x=1$ and $x=11$ (x is the time when sample was taken, in this case it is an hour bar on the EURUSD time series). So current time is $x=11$ and the value of it is known and it is the last known value. Those values x are result of tests which allows to make

conclusion that in this time series $k=11$ is optimal value of a time frame in which regression model should be tested. Current real value equals 1,2813. Value of linear model equals 1.2819. Value at the next step is unknown, but basing on a model this value can be predicted. Prediction for this step is 1,2821. Based on this prediction strategy it is advancing to a long position, because the predicted value is higher than the current real value, therefore growth is assumed. After the real y_{12} value is disclosed (which was unknown at the point of prediction) the decision would lead to failure, due to the real value of y_{12} being unexpectedly less than the value of y_{11} .

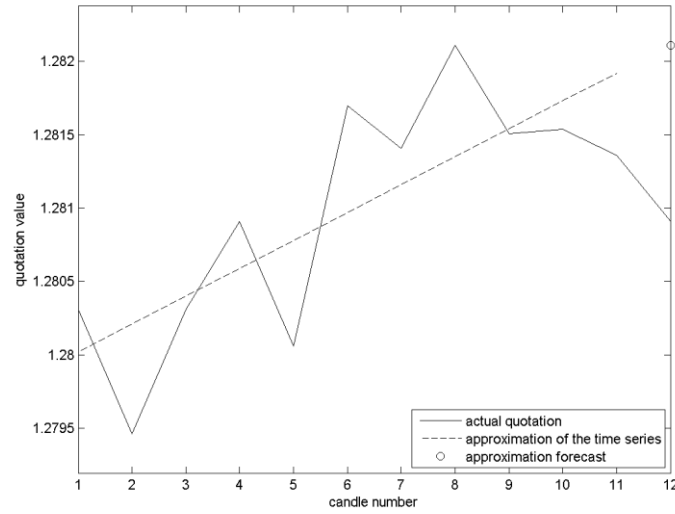


Fig. 1. Graphical representation of the flowchart strategy *regIt* on currency pairs market (Forex)

Formally, a polynomial model y_i^m describing values y_i , $i=1, \dots, k$ in sampling points x_i , $i=1, \dots, k$ can be represented by n degree polynomial:

$$y_i^m = a_n x_i^n + a_{n+1} x_i^{n-1} + \dots + a_1 x_i + a_0, \quad (1)$$

for $i=1, \dots, k$.

Vector $A=\{a_0, \dots, a_n\}$ selection is typically carried out using the two hundred year old, least squares method to minimize the sum of a differences between y_i and y_i^m for $i=1, \dots, k$.

The parameters of this strategy are: n degree of polynomial and size of time period k . Prediction is made using a vector of factors A , calculated in a previous step and shifted by a step to determine the value of y_{k+1}^m which is after the range of a sampled data.

Linear regression was determined using built-in MATLAB functions *polyfit* and *polyval*. Function *polyfit*, based on input data, is performing an interpolation and returning the factors of requested n degree polynomial. Function *polyval*, based on a given vec-

tor of a polynomial factors (form the *polyfit* function), is approximating result vector Y for a provided vector X.

Investment advice is made from a simple set of logical rules:

$$\text{if } \hat{y}_{i+1} > y_i \text{ and } a_1 > 0, \text{ then open long position} \quad (2)$$

$$\text{if } \hat{y}_{i+1} < y_i \text{ and } a_1 < 0, \text{ then open short position} \quad (3)$$

In those rules, factor a_1 is a factor of polynomial vector A placed by a linear value of the independent variable x . The second conditions, $a_1 > 0$ or $a_1 < 0$, were added to increase strategy efficiency. This new condition a_1 is a part of authors input into strategy based on the regression. The first condition stems from the obvious investor belief - if the model predicts an increase of value, long position is opened, as increase of price will give the profit.

The next strategy (*reg1tmetlife1*) is using the same prediction rule, the finance market is changed there to the investment fund. On an investment fund, short positions are not available. Third degree polynomial ($n=3$) was used in regression. Degree $n=3$ of polynomial can be a little surprise comparing to quiet efficient linear model for previous time series. Value of the param was set during attempts to find an optimum value (described as highest value of accumulated profit).

This strategy is trying to predict an investment funds value, which is a significantly slower time series than currency pair time series. The whole idea of investing in those markets is different than investing in currency or contract markets. What is particularly important to note is the fact that the value of investment fund is set after the close of a day. The results from the *reg1tmetlife1* strategy were used to build next script, *reg2tmetlife*, which is an extension on this method. Multiple funds from the Metlife family were put on the prediction. And only the best one from them is given as transaction advice.

According to the policy of investment funds, depositing funds is free of charge. In contrast to this, however, a large fee is added when investors want to withdraw their funds. This fee is often too costly that it may hinder the strategies ability to turn profit and, in some cases, could even result in some loss. Investment funds allows investors the freedom to move assets within one funds family, free of charge. When all considered investment funds are predicting the decrease of a price, to minimize loss the investor can move all of their assets to fund with the smallest drop or withdraw them to another fund within current funds family.

3 Investment Strategy Tests

Each of the implemented investment strategies were taken onto parameters optimization process. The params to optimize were the steps back used to predict value, and the degree of polynomial used within regression. The optimization objective function was to maximize the gained profit for one step forward prediction. Charges like swap (on financial markets) and costs of fund management were ignored during those tests. Compared to the results, those cost don't hold any significant influence on the final

result. It should be mentioned that for Forex strategies cost (difference between cost of selling and buying currency, called spread) has huge impact on a result. This has therefore been included in those strategies.

The first strategy optimization was carried out by searching the entire parameters range $k \times n$ to give a result of time period $k=11$ and polynomial degree $n=1$. This means that linear regression could be an efficient method to predict finance market value, limited to this market at that time.

Figure 2 shows the result of this strategy for the optimal parameters. On the abscissa is the number of tested OHLC bars. While the ordinate shows the profit (and loss) value. Expressed in pips are the differences between purchases on an actual step, and sale price on a next step (predicted one).

To check if strategy would perform successfully for different sets of parameters, some additional tests were taken:

1. For the same time period, as in figure 2 and for the linear model, different values of k (number of steps back) were tested. Results are shown in figure 3.
2. For the static $k=11$, values of $n > 1$ were tested. Results are shown in figure 4.

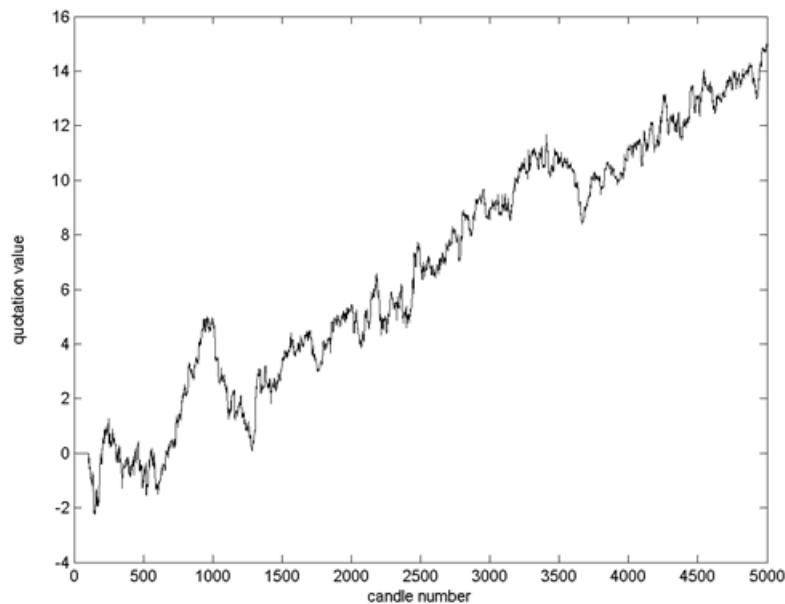


Fig. 2. The capital accumulation curve for the strategy *regIt* with optimal params on the test data.

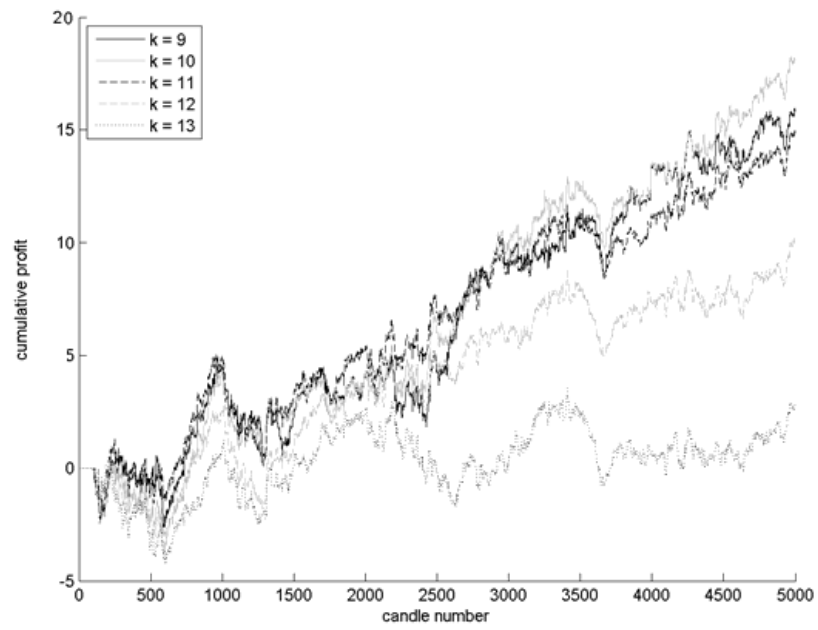


Fig. 3. The capital accumulation curve for various values of param k .

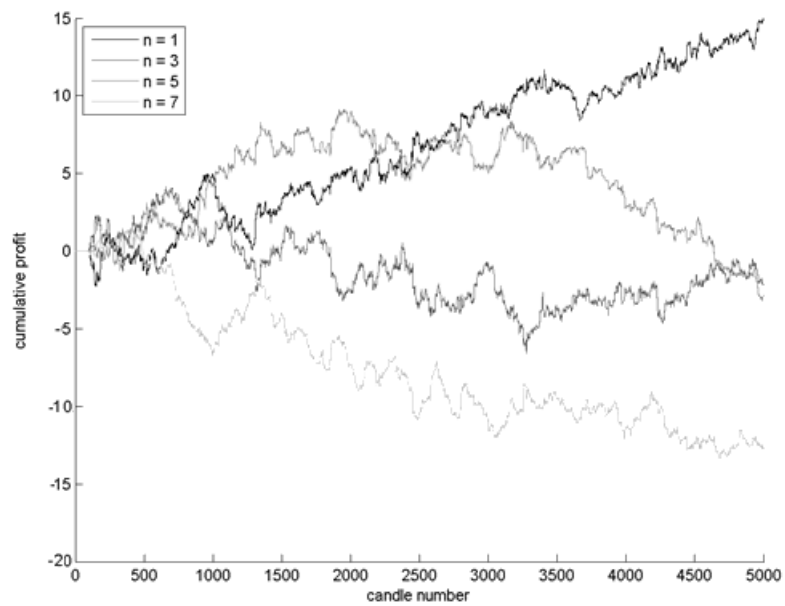


Fig. 4. The capital accumulation curve for various values of param n .

Results of a strategy with those parameter sets are not as promising as those of the most optimized one. However, for some curves on figure 3 and 4, long periods of growth could be found. This shows that some machine learning modifications should be applied to find the value of params n and k . This improved adaptive strategy was tested on a time series of the investment funds.

Figure 5. The capital accumulation curve for the strategy *reg1tmetlife1* with optimal params on the test data.

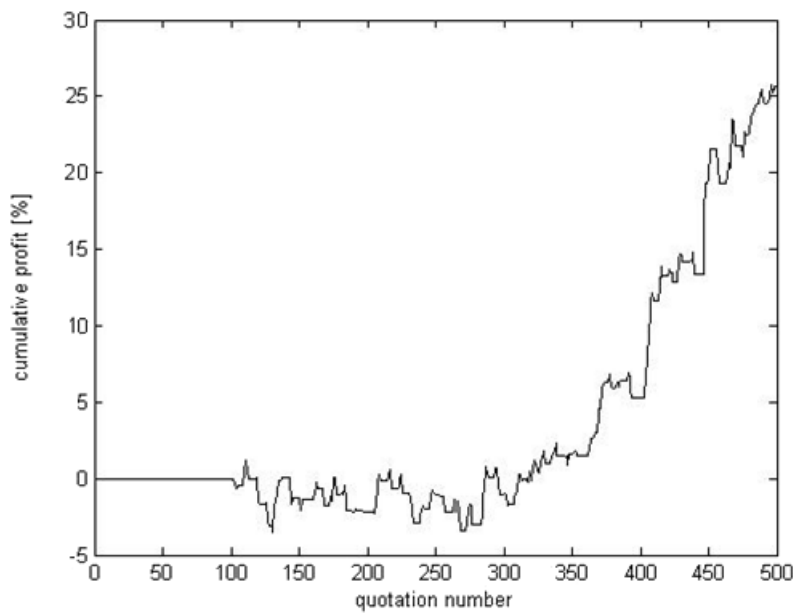


Fig. 5. The capital accumulation curve for the strategy *reg1tmetlife1* with optimal params on the test data

When regression strategy is used on investment funds market, optimal params are also expected to change. For the same time period $k=11$, better results were given by third degree polynomial. Small market inertia was also an influential factor when generating results, seen in figure 5.

Considered strategy opened only 215 long positions using approximately 400 samples (about 18 months). This equates to transactions being taken every two days on average. This would appear to be the underlying cause for differences between this chart and the chart for Forex market. Profits are shown as a relative values (relative to fund price at the day of purchase). Final profit from, seen in figure 5, could be interpreted as 26% after 500 days.

Next strategy implements switching between investment funds within one funds family (tested on the Metlife case). The optimal parameters for this strategy are $k=10$ and $n=4$. This strategy performed measurably well and produced very strong results.

Figure 6 shows the capital accumulation curves for each of the tested investment funds. It is clear that some of the funds resulted in loss, while those generating the highest profit are also characterized by significant capital drops. Results from spreading investment funds evenly to all funds (creating a fund portfolio consisting of all the funds of equal participation) are shown in figure 7. Experiments indicated that the diversification of funds between all investment funds within a family allows you to achieve around 3% profit after 400 days. This is not a satisfying result. Diversification of funds allows an investor to minimize investment risk. As shown in this study, all capital loss of individual funds are compensated by profits from other funds.

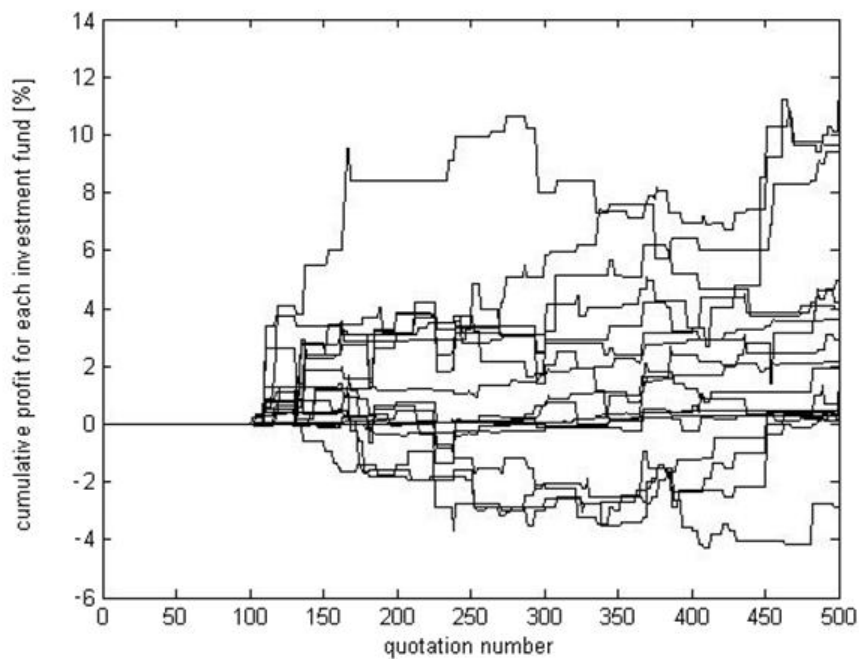


Fig. 6. The capital accumulation curves for each of the Metlife investment fund.

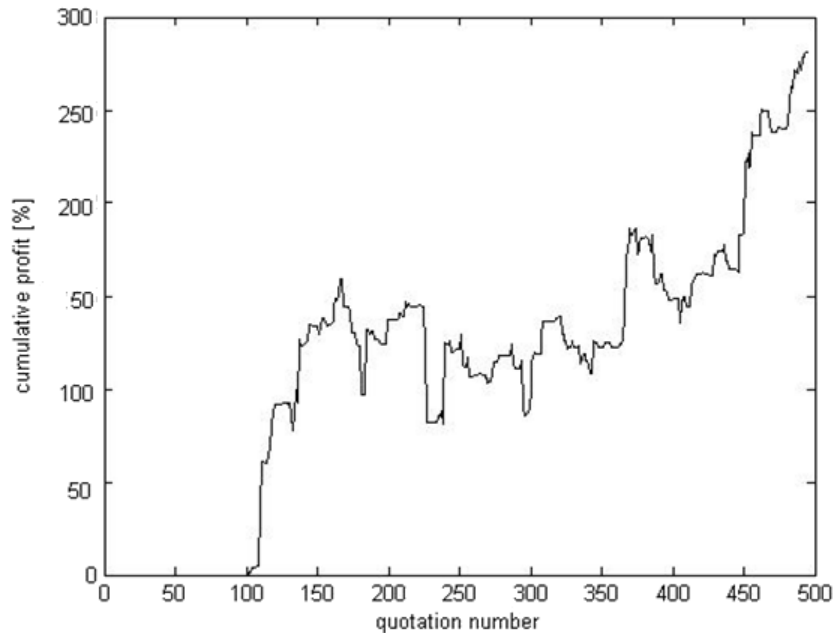


Fig. 7. The capital accumulation curve for the portfolio of equal participation of Metlife investment funds.

The strategy depicted in figure 8 is completely different to previous ones; transaction is made only on investment funds which gain a profit in a last step and, if there wasn't one matching to this rule, no transaction was taken. If there was more than one with high profit funds, it was spread equally between them. During these tests, there were multiple instances where none of the investment funds had growth in the last step, there were also multiple cases where many of them had growth. After adding these improvements, results are sensational and encourage the implementation of this strategy with a substantial degree of confidence. Difficult access to funds data published in various frequencies and with different delay, resulting in difficult downloading and processing, can be an obstacle to implementation.

It should also be noted that the described strategies do not include fund management costs which, depending on the individual case, can be up to 5% of its value on an annual basis.

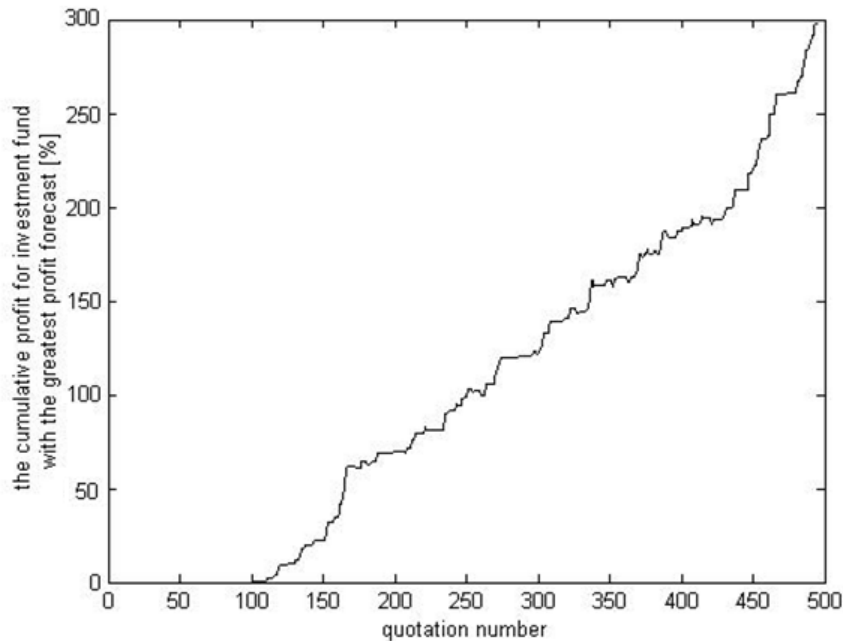


Fig. 8. The capital accumulation curve for the portfolio of single Metlife investment fund with the highest predicted profit.

4 Conclusions

Summing up the results from this paper, it can clearly be seen that the fund market is an untapped investment instrument. Ratio between the risk and the profit is usually measured by a few different criteria, ex. Sharpe or Calmar ratio. On the Figure 8 there is almost none of a dropdown and the highest of them is 5,46%. From over 200 of transactions made the best single gain was 11,29% and average profit was 0,69%. Not only does it offer long-term gain as an intermediate alternative between the bonds or the deposit and stock market investments (both in profits and risk), but it can also be used as a full-fledged speculative trading instrument. The use of automated trading strategies, although conceptual, stands in apparent contradiction with the philosophy of investment funds, but unexpectedly brings surprisingly good results.

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